

Chromosomes, genes, and mitosis

Genes can be described as 'units of inheritance' because each gene controls the development of one or a set of hereditary characteristics. Genes are not confined to controlling characteristics like eye colour or the shape of a nose. Each cell of a human body contains tens of thousands of genes and they are responsible for the development of all the tissues and organs as well as external features.

What are genes made of and where are they situated in an organism? Genes are made of complex chemicals described in Units 71 and 72, and they form part of objects called chromosomes, which make up the bulk of a cell nucleus.

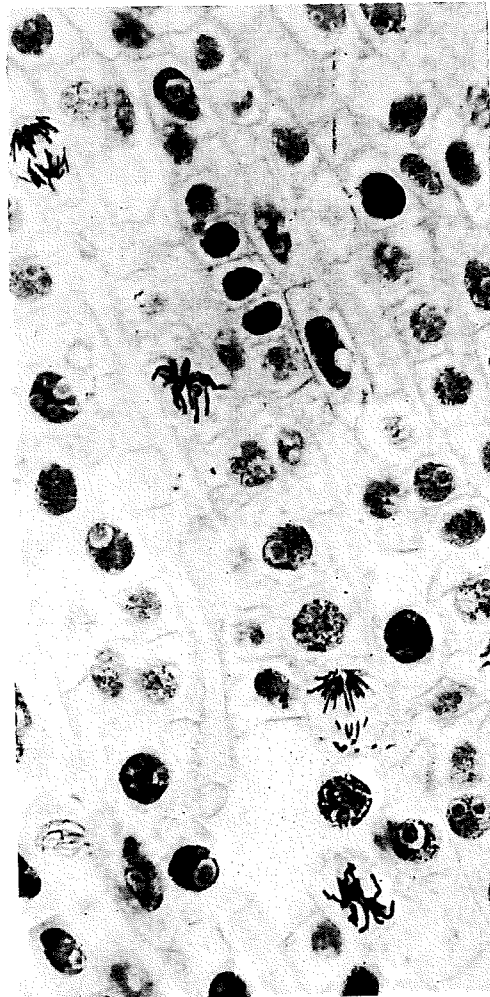
Humans and nearly all other multicellular organisms begin life as a single cell: a fertilized egg cell or zygote. A human zygote contains 46 chromosomes and between them they contain all the instructions for building a complete human being.

A human body is built from a zygote by cell division of a type called mitosis. During mitosis the zygote divides in such a way that the two daughter cells produced from it have the same number of

chromosomes as the zygote itself (Fig. 1 describes how this happens). Unit 71 describes how, just before a cell divides by mitosis, its chromosomes duplicate themselves, producing two identical sets of chromosomes. At mitosis one set of chromosomes passes into each daughter cell. Then these two cells divide by mitosis producing more cells which do the same so that, as the body grows, every cell receives 46 chromosomes which are an exact copy of the chromosomes in the original zygote, complete with a set of building instructions in the form of genes.

As growth proceeds instructions in the genes ensure that cells become tissues, that tissues become organs and, eventually, that individual features such as eye and hair colour appear.

Mitosis does not stop when the body reaches adult size. Mitosis in the epidermis produces cells which replace those worn away from the surface of the skin; it is responsible for producing new red and white blood cells in the bone marrow; it produces growth of hair and finger-nails; and it creates new cells to repair damaged tissues.



Chromosomes in the dividing cells from the tip of an onion root. Using the Figure opposite as a guide, identify the stages of mitosis.

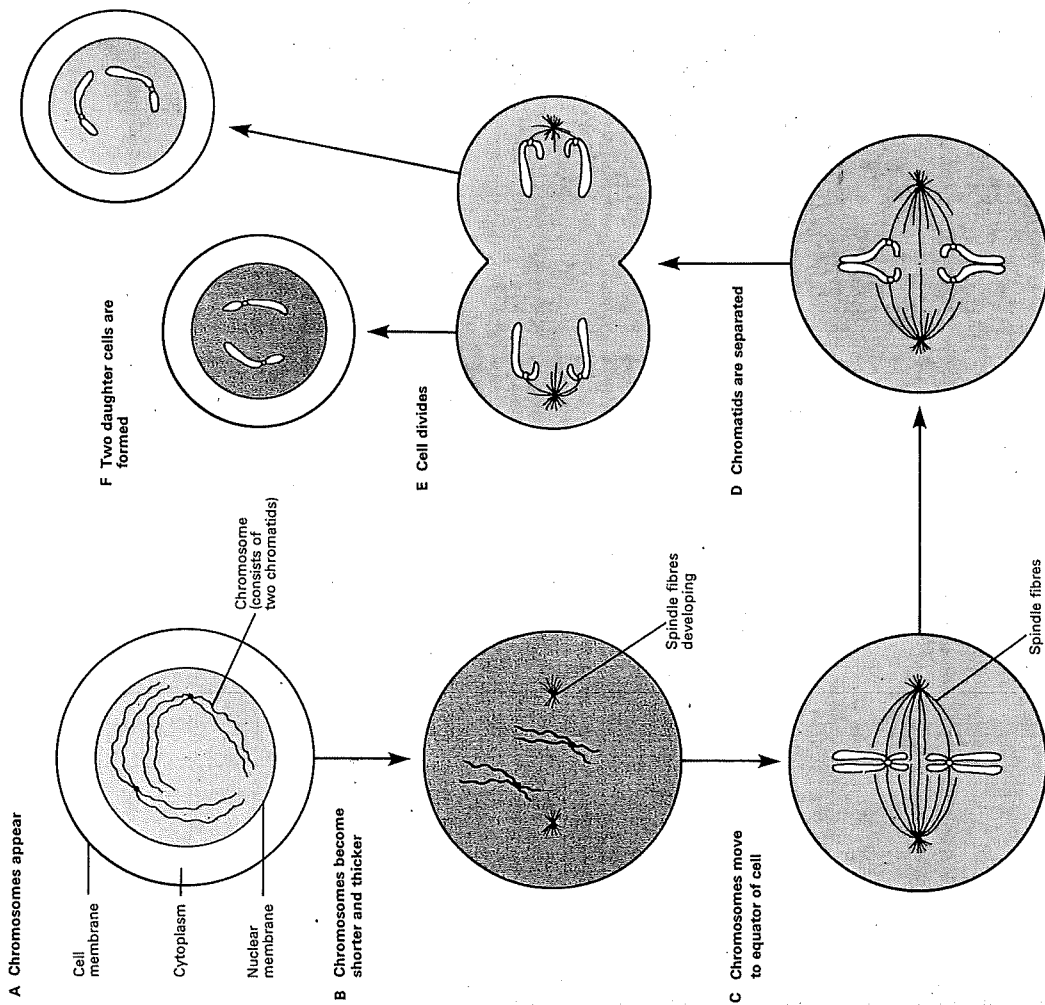


Fig. 1. Cell division by mitosis produces cells with the same number of chromosomes as the parent cell. Chromosomes are only visible when a cell is dividing. **A** and **B** Chromosomes first appear as long, fine double threads which become shorter and thicker. **C** The chromosomes move to the middle (equator) of the cell where they become attached to fine fibres called the

spindle. **D** Each chromosome separates into two parts which move to opposite ends of the cell. They are probably pulled apart by contracting ends of the spindle fibres. **E** and **F** The cell divides, separating the two groups of chromosomes which then form a nucleus in each daughter cell.

Gametes, meiosis, and sex chromosomes

Formation of gametes by meiosis

Humans have 46 chromosomes in every body cell. But if gametes (sperms and ova) had this number, fertilization would produce young with 92 chromosomes in their cells, and the number would double every time reproduction occurred.

This does not happen because gametes are produced by a type of cell division called meiosis. Unlike mitosis, described in Unit 67, meiosis produces four daughter cells with *half* the normal number of chromosomes. For this reason meiosis is called **reduction division**.

Human sperms and ova have 23 chromosomes. This is called the **haploid number** of chromosomes, and the normal or **diploid number** of 46 chromosomes is restored at fertilization. These 46 chromosomes are, in fact, 23 pairs. In technical terms they are called **homologous pairs**, partly because one member of each pair is the same shape and size as the other. Sex chromosomes, described below, can be an exception to this rule. The description of meiosis in Figure 2 should be studied before reading any further.

Genes and homologous chromosomes

The movement of chromosomes during meiosis and fertilization is clear evidence that they contain the units of heredity, called **genes**.

- Unit 70 explains that genes occur in pairs. During meiosis chromosomes form homologous pairs.
- Gene pairs separate at gamete formation (meiosis) and one member of each pair enters each gamete. Homologous chromosomes behave in exactly the same way during meiosis.
- Gene pairs and homologous pairs of chromosomes are restored at fertilization.

Sex chromosomes

The 23 pairs of chromosomes in human cells include a pair of sex chromosomes, so-called because they determine whether a zygote develops into a male or a female. There are two types of sex chromosome: the X and the Y types. Y chromosomes are smaller than X chromosomes.

A zygote containing a pair of X chromosomes develops into a female, but a zygote with an X and Y chromosome develops into a male. Figure 1 explains how the two types of zygote are produced.

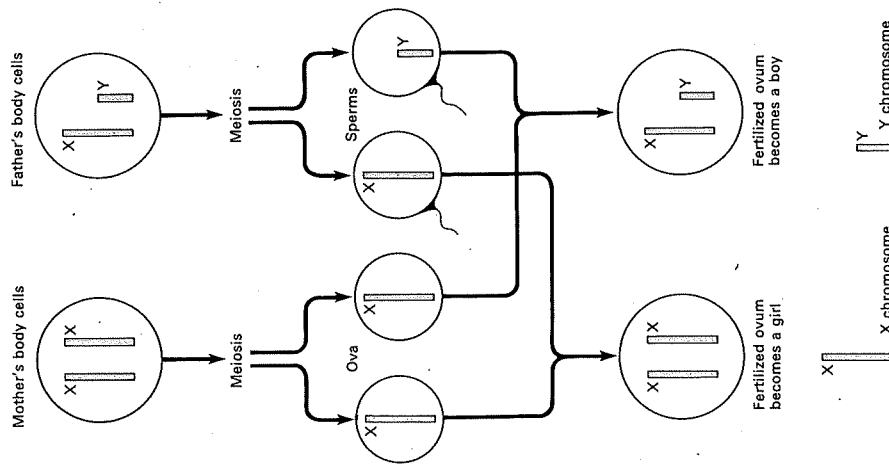


Fig. 1. Sex chromosomes determine the sex of a developing embryo. There are X and Y sex chromosomes. Females have two X chromosomes in their cells and males have an X and a Y chromosome in each cell. The diagram shows how X and Y chromosomes separate at meiosis and come together again at fertilization.

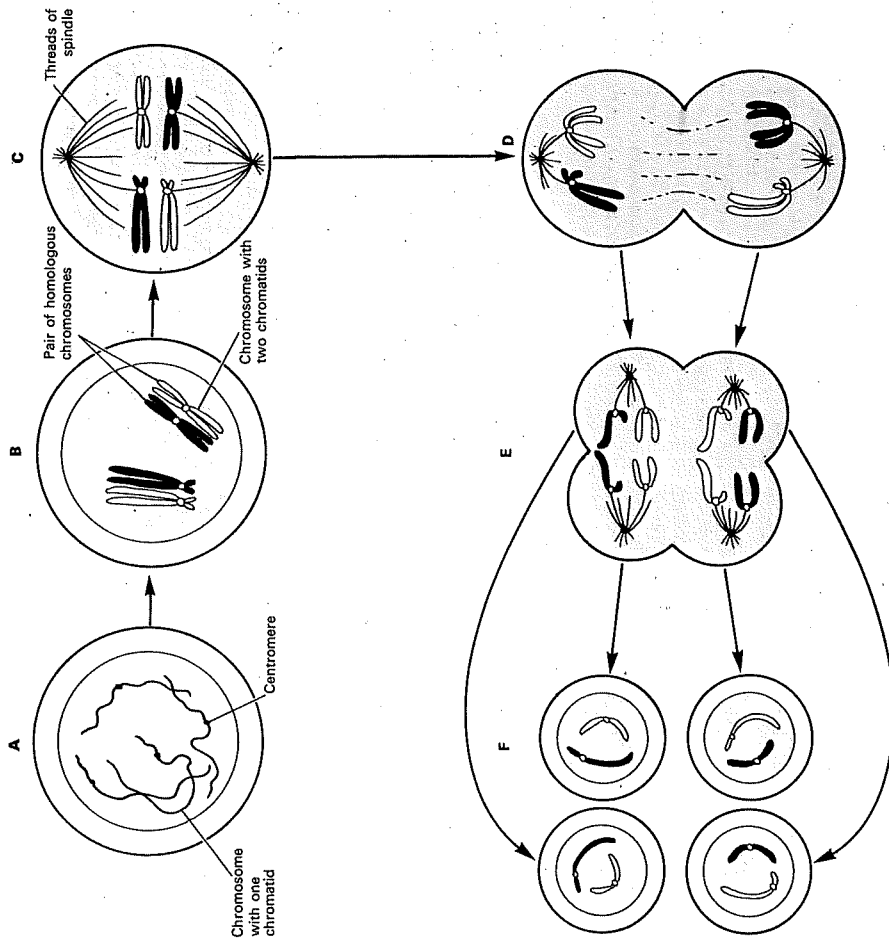


Fig. 2. Cell division by meiosis produces cells with half the number of chromosomes as the parent cell. Meiosis occurs in the testes and ovaries producing sperms and ova with 23 chromosomes each. **A** Chromosomes become shorter and thicker. **B** Chromosomes form homologous pairs. (One of each pair is drawn in black to distinguish it from its partner.) Each chromosome forms a second strand. **C** The homologous pairs become arranged along the equator of the cell, attached to spindle fibres. **D** The members of each homologous pair of

chromosomes separate and move in opposite directions. The cell begins to divide into two. **E** Each chromosome splits into two parts which move in opposite directions. **F** The cell divides into four parts each containing half the number of chromosomes possessed by the original cell. In the testis the four new cells develop into sperms. In the ovary only one of the four develops into an ovum. The remaining three are very small and soon disintegrate.

Some technical terms used in genetics

Homozygous and heterozygous gene pairs

Genes operate in pairs, and each pair controls one or a set of hereditary characteristics. When the members of a particular gene pair are identical the pair is said to be **homozygous**. A gene pair can be homozygous dominant (AA) or homozygous recessive (aa).

The members of a gene pair can be different: one dominant and the other recessive (Aa). In this case the pair is said to be **heterozygous**.

The locus and alleles of a gene

The position of a gene on a chromosome is called the **locus** of that gene. Unit 68 explains that chromosomes occur in pairs, called homologous chromosomes. Each pair of homologous chromosomes holds a particular set of genes, arranged so that the members of each gene pair occupy the same locus on both homologous chromosomes.

Genes which occupy the same locus on homologous chromosomes are said to be **alleles** of each other. Inheritance of blood groups, for example, is controlled by three alleles: the genes A, B, and O. But, since genes operate in pairs, only two of these alleles can be present at the same time in an individual's cells. Gene O is recessive to both A and B, but A and B are said to be **co-dominant** because neither can 'dominate' the other. Table 5 shows how the different combinations of these alleles produce the various blood groups.

Genotype and phenotype

The set of genes possessed by an organism is referred to as the **genotype** of that organism. The word genotype can also refer to one particular set of genes, such as those controlling eye colour. In contrast, the word **phenotype** refers to the actual visible physical characteristics of an organism's body.

For example, two organisms can have different genotypes but the same phenotype. Wing length in *Drosophila* is controlled by genes which can be represented by A and a. Genotypes AA and Aa are different but produce flies with the same phenotype: normal (long) wings. This is because A is dominant. On the other hand two organisms with the same genotype can have different phenotypes. 'Identical' twins, for instance, have exactly the same genotype because they developed from the same zygote. As the

twins grow, however, interaction between these genes and their environment can result in different phenotypes. In particular they may differ in height, weight, and intelligence, especially if they are separated at an early age and reared in different environments.

Mutation

A mutation is a sudden change in a gene or chromosome which alters the way in which it controls development. Mutations are usually harmful, but are quite rare. Mutations occur more frequently in organisms exposed to X-rays, gamma rays, and nuclear radiation. The most important mutations are those in cells which produce gametes (sex cells) because these mutations can pass from parents to young.

Albinos, for example, result from a mutation in the genes which control skin colour (see photograph opposite). Haemophilia is a disease which results from mutation in a gene which controls the formation of the blood-clotting mechanism. The blood of a haemophiliac fails to clot in wounds. Inheritance of haemophilia is sex-linked.

Sex-linked inheritance

Certain types of haemophilia and colour blindness are examples of sex-linked characteristics, so-called because they are more likely to occur in males than females. There are several reasons why this happens. The genes for haemophilia and colour blindness occur on X chromosomes, and both genes are recessive to a corresponding dominant allele for normal blood clotting and colour vision. Females have two X chromosomes. Consequently a gene for colour blindness or haemophilia on one X chromosome will, in most cases, be masked by a dominant allele on the other X chromosome. Very few females have the recessive genes on both X chromosomes.

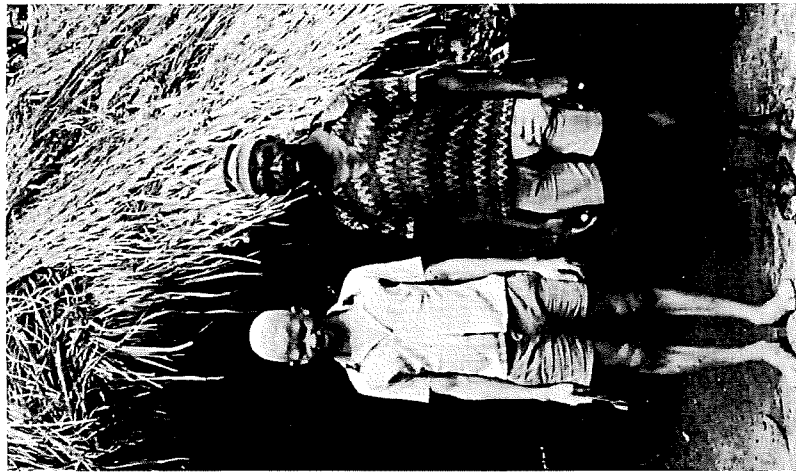
Males have one X and one Y chromosome in their cells. The Y chromosome is small and lacks both blood-clotting and colour-vision genes. If therefore, a male has received from his mother an X chromosome with a gene for colour blindness or haemophilia he has no corresponding dominant allele and will suffer from the disorder.

Figure 1 illustrates the passage of sex-linked genes from parents to young.

Table 5. Inheritance of blood groups.

Genes	Blood groups
OO	O
AA	A
AO	A
BB	B
BO	B
AB	AB

O is recessive to A and B but A and B are co-dominant—neither is dominant to the other.



An albino and a normal dark-skinned African. Albinism results from a mutation in the genes which control skin coloration. The skin of an albino is pink and almost transparent and is quickly damaged by exposure to sunlight. The mutation is recessive.

Key
 X X chromosome with normal blood-clotting gene
 $\bar{X}$ X chromosome with haemophilia
 Y Y chromosome

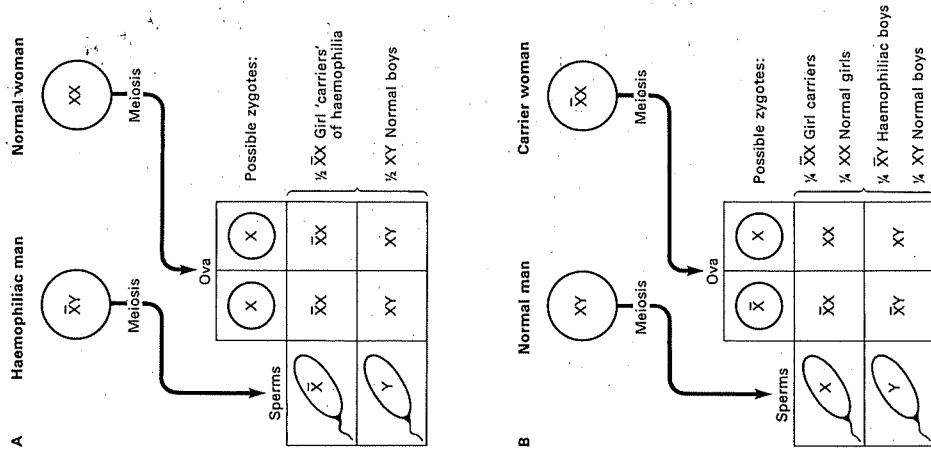


Fig. 1. Inheritance of haemophilia is sex-linked. Haemophilia genes pass from a father to his daughters, but from a mother to her sons and daughters. Sons who receive the gene are haemophiliacs, daughters are carriers unless, very rarely, they receive two recessive genes.

DNA and the genetic code

Genes are the instructions which direct the growth and development of organisms. The popular name for these instructions is the **genetic code**. What is the genetic code made of and how are its instructions obeyed? This Unit and the next try to answer these questions.

In 1944 it was shown that genes are made of a chemical called **deoxyribonucleic acid**, or **DNA**. In the early 1950s the structure of DNA was discovered and this led to an understanding of the genetic code.

Structure of DNA

DNA is a long, thread-like molecule similar in shape to a rope ladder twisted into a spiral (Fig. 1). The upright sides of the ladder are made of alternate sugar and phosphate molecules joined into a chain. The rungs of the ladder are made of chemical bases, of which there are four types: adenine, guanine, cytosine, and thymine. For simplicity these bases are usually referred to by their initial letters A, G, C, and T (Fig. 2).

The genetic code

The genetic code is formed by the sequence in which these four chemical bases are arranged along the length of a DNA molecule. Put another way, if the genetic code were a set of written rather than chemical instructions, the letters A, G, C, and T would be the alphabet in which the 'words' of the code were written.

In fact, the code is written in words of only three letters each; that is, it consists of groups of three adjacent bases. For example, the base sequence AAT along one strand is one word, CAA is another, and so forth. The number of different three-letter words which can be composed out of the four letters A, G, C, and T is 64.

It is now known that these three-letter words (groups of three bases) are the instructions to a cell telling it how to make protein molecules out of amino acids. The sequence AAT (adenine, adenine, thymine) in the code tells a cell how to make the amino acid leucine, and CAA (cytosine, adenine, adenine) tells a cell how to make the amino acid valine.

A single gene consists of a thousand or more

groups of three bases. Together these groups represent instructions to make one complete protein molecule of a specific type. A cell 'obeys' these instructions by putting together the correct amino acids in the exact sequence directed by the gene. The next Unit explains how this happens.

DNA, therefore, controls the types of protein manufactured in cells. In this way DNA controls the structure of an organism because proteins are the building materials out of which cells, tissues, and organs are made. But equally important is the fact that DNA directs the chemical activities which take place in an organism, because some proteins are enzymes and enzymes control the speed and type of chemical reactions in cells.

Duplication of DNA during cell division

Unit 67 explains how cell division by mitosis produces daughter cells with exact copies of the chromosomes present in the original cell. Before mitosis can occur in a cell the DNA molecules in its chromosomes must be duplicated exactly. Eventually these duplicates form a second set of chromosomes.

Before describing DNA duplication it is necessary to look more closely at its structure. The rungs of the ladder-like DNA molecule are made up of chemical bases represented by A, G, C, and T. Each rung consists of two bases joined at the centre of the ladder (Fig. 2). There are millions of rungs in one DNA molecule but in every one, base A is *always* opposite base T, and base C is *always* opposite base G. This is known as the **pairing rule**.

When DNA duplicates, the base pairs in each rung separate and the molecule comes apart into two strands as if it had been unzipped up the middle (lower half of Fig. 2). The two single strands are now built up into double strands again using new bases supplied by the cell. The pairing rule ensures that the base T is always joined up with an A (or an A with a T) and the base C is always joined up with a G (or a G with a C) as shown in Figure 2. The result is two DNA molecules which are identical to the original.

Very rarely an error occurs during duplication of DNA and results in a slight change in the genetic code. This is one way in which mutations occur.

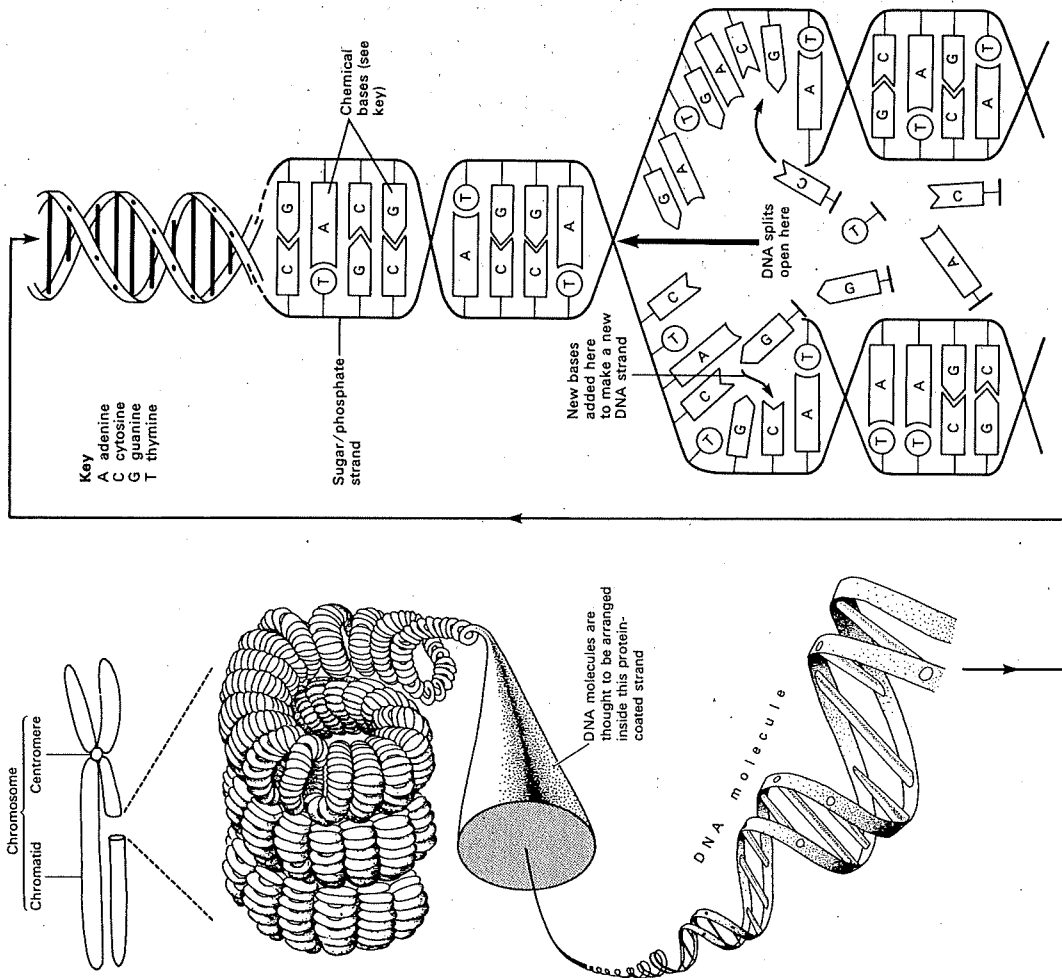


Fig. 1. Chromosomes and DNA. A chromosome consists of strands of protein which coil up as shown just before cell division. It is possible that DNA molecules run through the centre of these protein strands. A DNA molecule consists of two outer filaments with cross-pieces between them—like a ladder twisted into a spiral.

Fig. 2. Structure and duplication of DNA. The two outer strands of a DNA molecule are made of alternate sugar and phosphate molecules. The cross-pieces consist of chemical bases: A (adenine), G (guanine), C (cytosine), and T (thymine). A is always opposite T, and C is always opposite G. Duplication of DNA occurs just before cell division. The molecule splits into two strands and new chemical bases are added to these, building up a new double strand again. The result is two DNA molecules identical to the original one.

Transcribing the genetic code

The previous Unit explains that each gene on a DNA molecule carries instructions for making a particular type of protein. This Unit describes how these instructions are obeyed or, in technical terms, how the genetic code is transcribed. This Unit also explains how each cell transcribes only those genes concerned with its particular functions leaving the thousands of other genes in its chromosomes inactive.

Most of a cell's DNA is contained in the chromosomes in its nucleus, but proteins are made in the cytoplasm of a cell. Somehow the coded instructions for protein manufacture must pass from the nucleus to the cytoplasm and then be transcribed. This task is carried out by two types of a substance called ribonucleic acid, or RNA, which work in conjunction with microscopic granules in the cytoplasm called ribosomes. The two types are called messenger RNA and transfer RNA. Figure 1 and the following notes describe the commonly accepted theory of how this transcription happens.

A DNA molecule opens up

The region of DNA containing the gene for the protein required by the cell opens out, exposing the sequence of chemical bases with the necessary coded instructions (Fig. 1A).

Messenger RNA copies the code

The exposed single strand of DNA with the coded instructions is built up into a double strand again in a similar fashion to DNA duplication described in Unit 71, except that this time the new second strand is made of messenger RNA.

The main difference between RNA and DNA is that in RNA the base thymine is replaced by the base uracil, which is represented by the letter U. Consequently the messenger RNA strand is built up according to the pairing rule: A (adenine) opposite U (uracil), and G (guanine) opposite C (cytosine). In this way the messenger RNA molecule becomes an exact copy of the exposed gene's coded instructions on the DNA molecule.

Next, messenger RNA leaves the nucleus and moves into the cytoplasm where it becomes attached to a ribosome (Fig. 1B). This is where protein building takes place.

The cytoplasm of a cell contains millions of amino acids—the building blocks from which proteins are

made. The human body requires 20 different amino acids to build its proteins, and each type of protein molecule contains a different number of amino acids arranged in a different sequence.

Transfer RNA collects the amino acids

It is the function of transfer RNA molecules to move around the cytoplasm collecting the correct types of amino acid (Fig. 1C). Transfer RNA then transfers the amino acids to messenger RNA on a ribosome, where the amino acids are linked up in the correct sequence to make a protein molecule.

A transfer RNA molecule contains three chemical bases, and Unit 71 explains that a group of three such bases is the code for a particular amino acid. Each transfer RNA molecule picks up a single amino acid molecule of the type directed by its part of the code, and floats around the cytoplasm until it reaches a ribosome which has a strand of messenger RNA attached to it.

Amino acids are linked together

When a transfer RNA molecule (complete with its amino acid) arrives at a ribosome its three chemical bases join up with a corresponding set of three bases on the messenger RNA molecule, again using the pairing rule A opposite U and G opposite C. As this is happening the amino acid carried by the transfer RNA links up with amino acids on an adjacent transfer RNA molecule. Repetition of this process builds up a chain of amino acids until a complete protein molecule is formed (Fig. 1D).

Summary

Ribosomes, together with messenger and transfer RNA, form an assembly line for proteins, and DNA is the library where assembly instructions are stored.

Messenger RNA molecules visit the DNA library, copy the instructions for making a protein, and carry them to a ribosome in the cytoplasm. Transfer RNA molecules collect amino acids and transfer them to the ribosome. Here, the amino acids are fitted together according to the instructions built into the structure of the messenger RNA.

The sequence in which the amino acids are fitted together depends upon the sequence of bases in the messenger RNA, which depends upon the sequence of bases in the region of DNA which was copied.

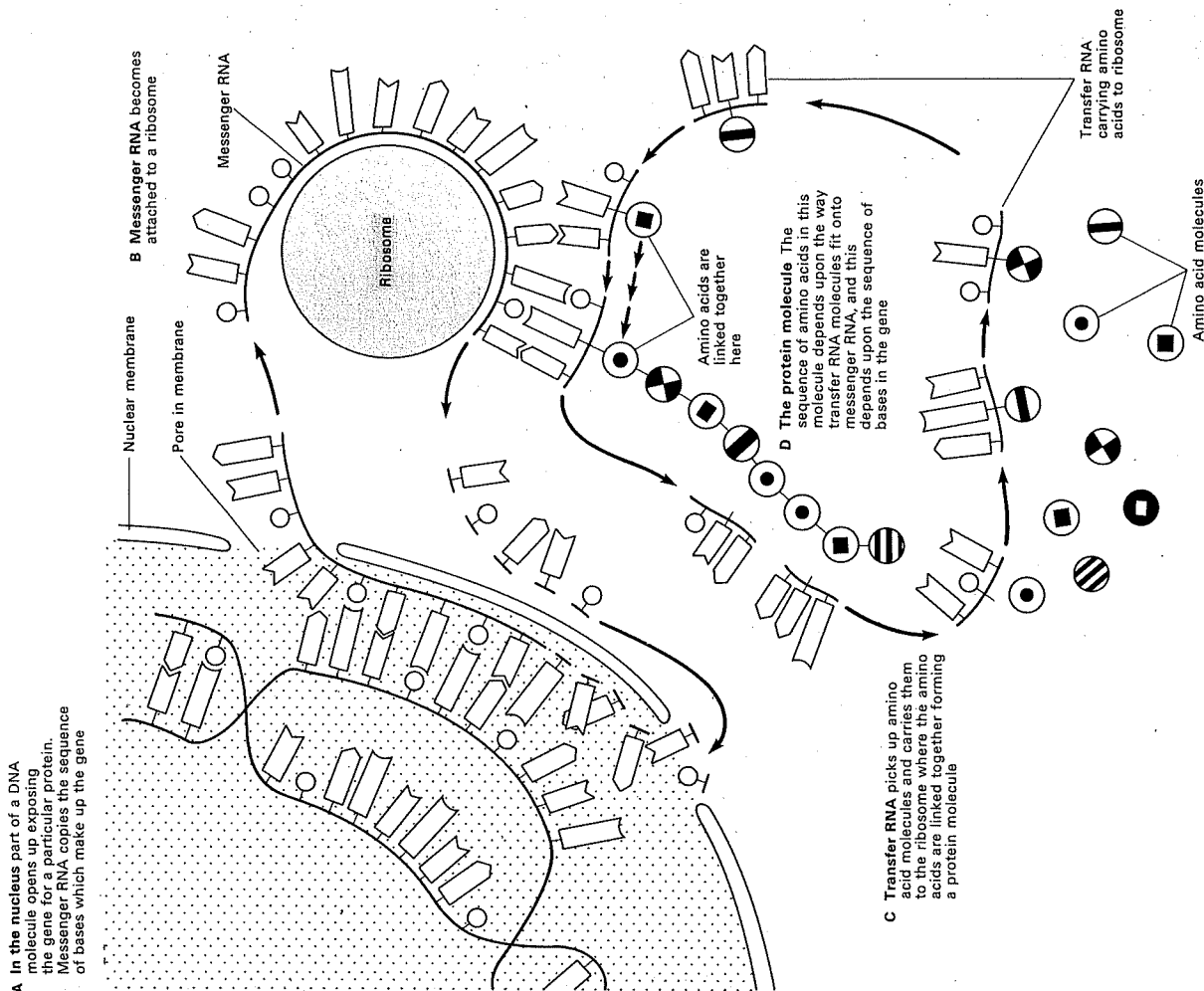


Fig. 1. Transcription of the genetic code.

DNA contains the instructions for growth and development in humans and all living things. Our DNA is packaged into chromosomes that contain all of our genes.

In summary

- DNA stands for (DeoxyriboNucleic Acid) which is made up of very long chains of chemical 'letters': Adenine (A), Guanine (G), Thymine (T) and Cytosine (C)
- DNA contains the instructions for our genes
- Genes are the instructions for making proteins. Proteins do the work within our cells and body
- In humans, most genes are arranged on chromosomes that are found in the nucleus of cells.

THE GENETIC BOOK OF LIFE

In humans genetic information, also known as our genome, can be described as the 'Book of Life'. This book can be thought of as being made up of two volumes, each volume of the book is given to a person by one of their parents (*Figure 1.1*).

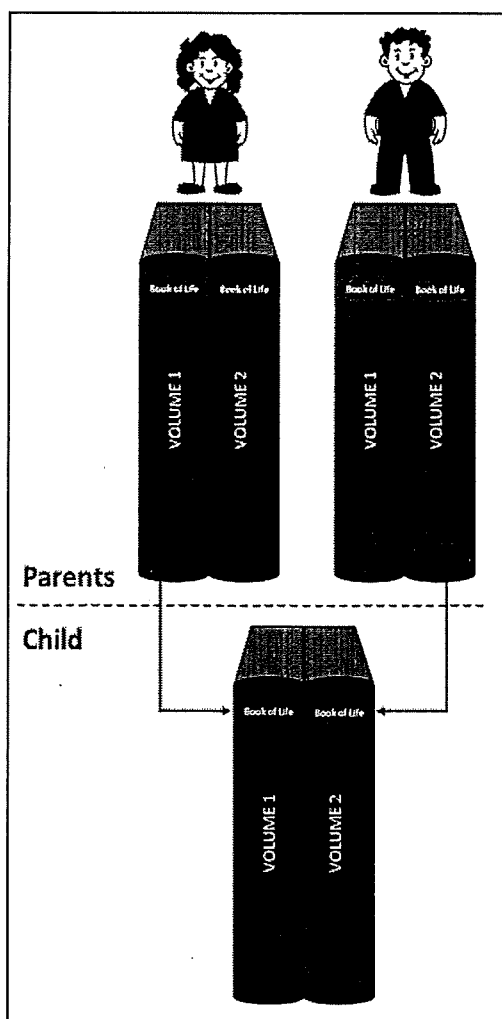


Figure 1.1: The human genome sometimes called the 'Book of Life'

Reading a person's genetic book of life (*Figure 1.2*):

- One volume of the book is inherited from the mother and the other from the father
- Both volumes contain 23 chapters each, equivalent to the 23 pairs of chromosomes in human body cells that contain genetic information
- The 23 chapters (chromosomes) are made up of a number of recipe pages (coding DNA or genes) and in-between (non-coding) pages of DNA
- Some of the chapters contain many pages while others only have a few. Some chromosomes are large and contain many thousands of genes and non-coding DNA while others are much smaller
- Genes are sections of DNA that code for the proteins our body needs to function
- In-between (non-coding) sections of DNA have various jobs, not all of which we understand
- Careful examination of the words within genes shows that all the words are made up of three letters (triplets) such as AGT, GGT, ACT, CAA etc.
- There are four letters used in the genetic book. They are A, T, C & G.

Just as reading the words on the page of a book allows an understanding of the author's message, the body is able to read the triplets in genes to make the protein needed for our cells to work.

Our cells don't need all the instructions all the time. Pages of our genetic book can be closed and then reopened when needed. Each type of cell can have different parts of the genetic book opened or shut because different cells do different jobs in our body. Which genes are turned on or off can be influenced by our diet, chemical exposure, exercise, ageing and messages from other genes in the body.

The Book of Life		Genetic Terminology
Volume 1 & 2	→	Genome
Chapters	→	Chromosomes
Pages	→	Genes
Words	→	Triplets
Letters	→	A, T, C, G
Spelling Mistake	→	Genetic Variant

Figure 1.2: Genetic terminology and the 'Book of Life'

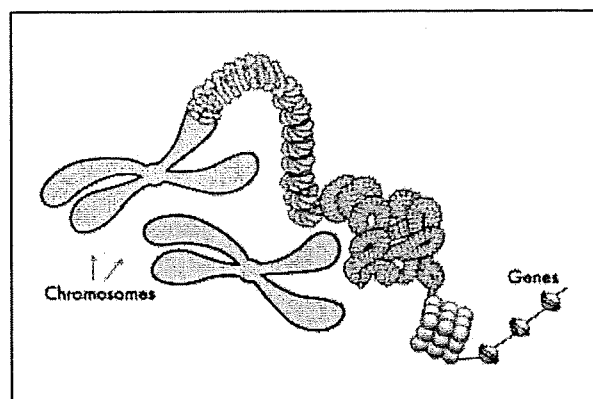


Figure 1.3: Chromosomes are like strings of genes

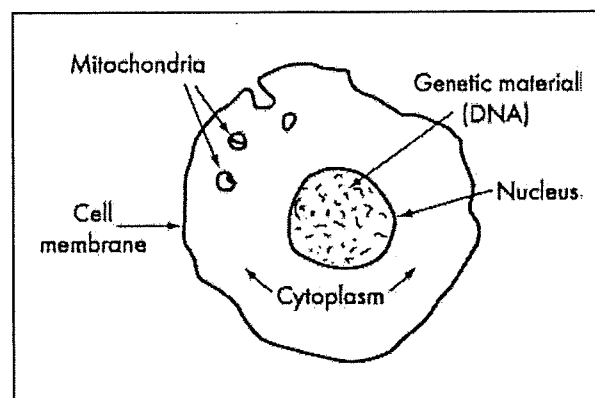


Figure 1.4: Diagram of a human cell

DNA, GENES AND CHROMOSOMES IN THE BODY

Our bodies are made up of millions of cells. Each cell contains a complete copy of a person's genetic book of life.

Chromosomes can be thought of as being made up of strings of genes (DNA that codes for proteins) with non-coding DNA between them. The chromosomes, including the genes, are made up of a chemical substance called DNA (DeoxyriboNucleic Acid).

The chromosomes are very long strands of DNA, coiled up like a ball of string as shown in Figure 1.3.

Chromosomes are found in the nucleus of all body cells except for red blood cells which have no nucleus and therefore do not contain chromosomes.

Another place in the cell where DNA is found is in very small compartments called **mitochondria** (the energy centres of the cell) that are found scattered outside the nucleus (Figure 1.4). The DNA in mitochondria is much smaller and has very little non-coding DNA.

Chromosomes

There are 46 chromosomes contained in the **nucleus** of body cells:

- Of these, 23 came from the mother's egg and 23 came from the father's sperm
- When the egg and the sperm join together at the time of conception, the first cell of the baby is formed. This cell is copied to make all of the cells of the baby
- The baby's body cells now have 46 chromosomes, made up of 23 pairs, just like the parents (Figure 1.5).

As we age and grow, our cells are continually dividing to form new cells.

During this division process, each of the long thin chromosomes coils up tightly, so that each of the 46 individual chromosomes in the nucleus become rod-shaped structures and can be seen when using a microscope.

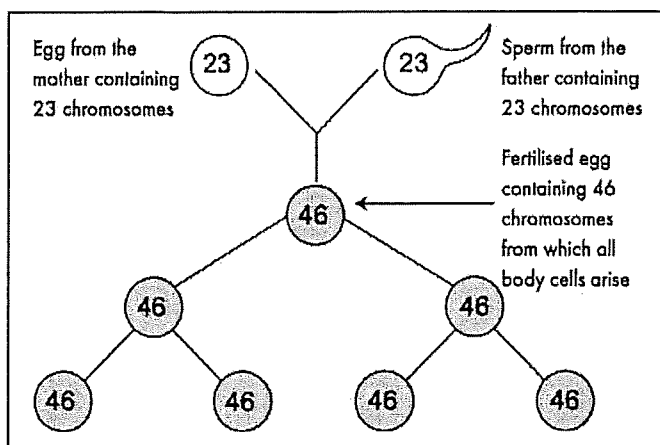


Figure 1.5: At conception the sperm and egg combine

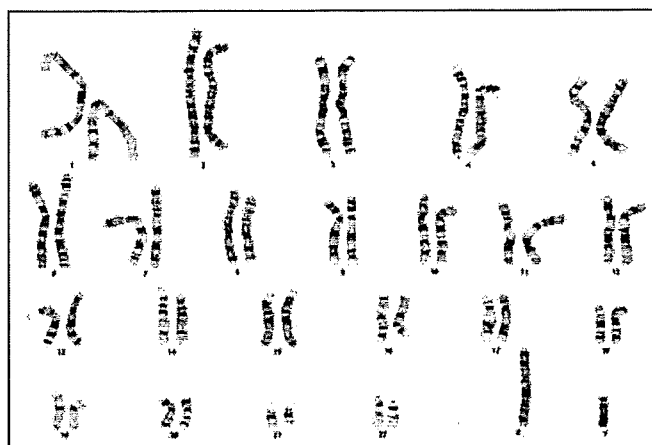


Figure 1.6: Chromosome picture (karyotype) from a male 46,XY.

In a genetic testing laboratory the chromosomes may be coloured (stained) with special dyes to produce distinctive banding patterns and lined up in size order. This produces what we call a **karyotype**. These patterns allow the laboratory to check the size and structure of each chromosome.

Figure 1.6 shows a banded chromosome karyotype where each chromosome has been numbered from the largest (chromosome number 1) to the smallest (chromosome number 22) and arranged in pairs in order of size. These numbered chromosomes are called **autosomes**.

There are two chromosomes that have been given the labels X and Y. These are the **sex chromosomes**. It is these sex chromosomes that determine whether the chromosomes have come from a male or a female.

In females, cells in the body have 46 chromosomes (44 autosomes plus two copies of the X chromosome). They are said to have a **46,XX karyotype**. Eggs (female reproductive cells) are different as they only contain half of the chromosomes (23 made up of 22 numbered chromosomes and an X chromosome).

In males, cells in the body have 46 chromosomes (44 autosomes plus an X and a Y chromosome). They are said to have a **46,XY karyotype**. Sperm (male reproductive cells) are different as they only contain half of the chromosomes (23 made up of 22 numbered chromosomes and an X chromosome or a Y chromosome).

Genes

The DNA making up each chromosome is usually coiled up tightly. If we imagine it stretched out, it would look like beads on a string (Figure 1.3):

- Each of these beads is called a **gene**
- Each gene is an instruction for a specific protein
- Thousands of genes make up each chromosome
- Between the genes are segments of non-coding DNA.

Since the chromosomes come in pairs, there are also two copies of each of the genes. The exception to this rule applies to the genes carried on the sex chromosomes, X and Y.

Since men have only one copy of the X chromosome, they have only one copy of all the genes carried on the X chromosome. Women have two copies of the X chromosome in their cells and so they have two copies of all the genes carried on the X chromosome.

To adjust for the fact that women have two X chromosomes with lots of genes while men have only one, one of the woman's X chromosomes is **switched off** or inactivated in each of their cells.

There are very few genes on the Y chromosome and their role is mainly to make a person male, so they are not needed in female cells.

DNA

There are over 20,000 genes found in the DNA of each person. Each gene has its own specific location on a chromosome or on the mitochondrial DNA and the genes (coding DNA) plus the non-coding DNA make up that person's **genome**.

- The DNA code is made up of very long chains of four basic building blocks (*nucleotide bases*) called **Adenine (A)**, **Guanine (G)**, **Thymine (T)** and **Cytosine (C)**
- A chromosome consists of two of these DNA chains running in opposite directions. The bases pair up to form the rungs of a ladder twisted to form a double helix (*Figures 1.7 & 1.8*)
- Pairing of the bases follows a pattern where base A can only pair with base T and base G can only pair with base C. Roughly three billion of these base pairs of DNA make up the human genome
- Our DNA code is made up of a combination of three of these four chemical 'letters' called a triplet.
 - ◊ Each three-letter word (*triplet*) tells the cell to produce a particular amino acid, the building blocks of proteins
 - ◊ The sequence of three-letter words in the gene enables the cells to assemble the amino acids in the correct order to make up a protein
 - ◊ Only about 2% of the entire DNA in the human cell is made up of genes that contain the information codes for making proteins
 - ◊ The remaining 98% of DNA does not contain the information for proteins and used to be called **junk** DNA. This non-coding DNA separates genes from each other along the chromosomes and there is increasing evidence that it has a role in turning genes **on** and **off**. This non-coding DNA therefore has a control function within the genome.

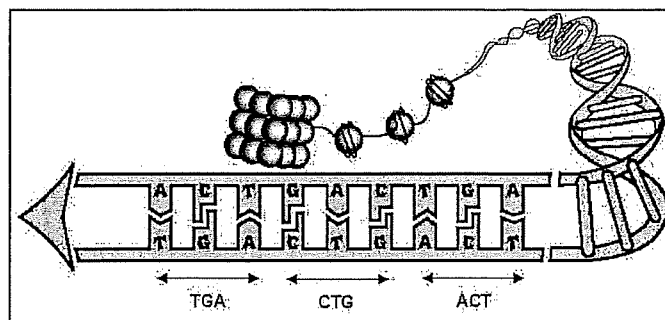


Figure 1.7: The DNA bases pair up to make genes

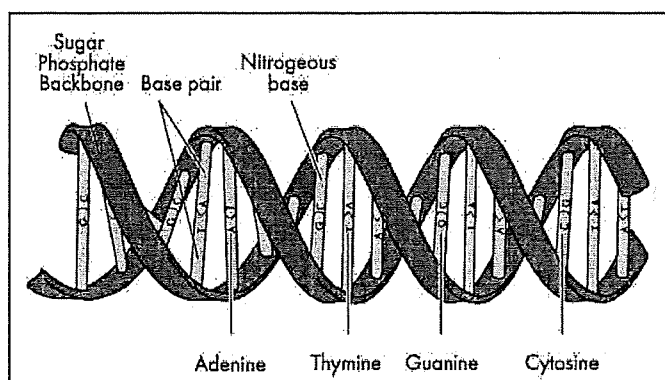


Figure 1.8: The DNA helix

DNA VARIATIONS

We all have small variations in our genetic code. That is why we are all unique. Even identical twins have some variations in their DNA by the time they are born. Because we inherit our genes from our parents, members of the same family share their DNA including its variations.

There may be changes in the sequence of letters in the gene message; nucleotide base/s (A, G, T or C) can be missing (called a deletion) or base/s can be added (called an insertion) and these can be of one or many DNA bases.

Variations in the code can occur during our life for a variety of reasons including exposure to radiation, certain chemicals or by chance. Ageing is a common cause of genetic variation. Throughout our lives, our cells are continually being replaced.

Some variations in the genetic information do not seem to make any difference to the function of our cells. These types of DNA variations are quite common.

Other DNA variations can be associated with an increased chance of a health condition, for example diabetes or cancer.

Some DNA variations can mean the gene instruction is incorrect so a faulty protein is made or the control switch is changed. A variation in a gene that creates a fault is called a **pathogenic variant** or **mutation**.

A DNA mutation can cause a problem for one cell type but not another, since not all cells use all of the possible proteins.

When a DNA change causes a faulty protein in cells that need that protein, it usually results in disease symptoms that can sometimes be recognised as a genetic condition.

Since we have two copies of each gene, if one copy has a mutation and the other copy is working, then we may not develop any problems.

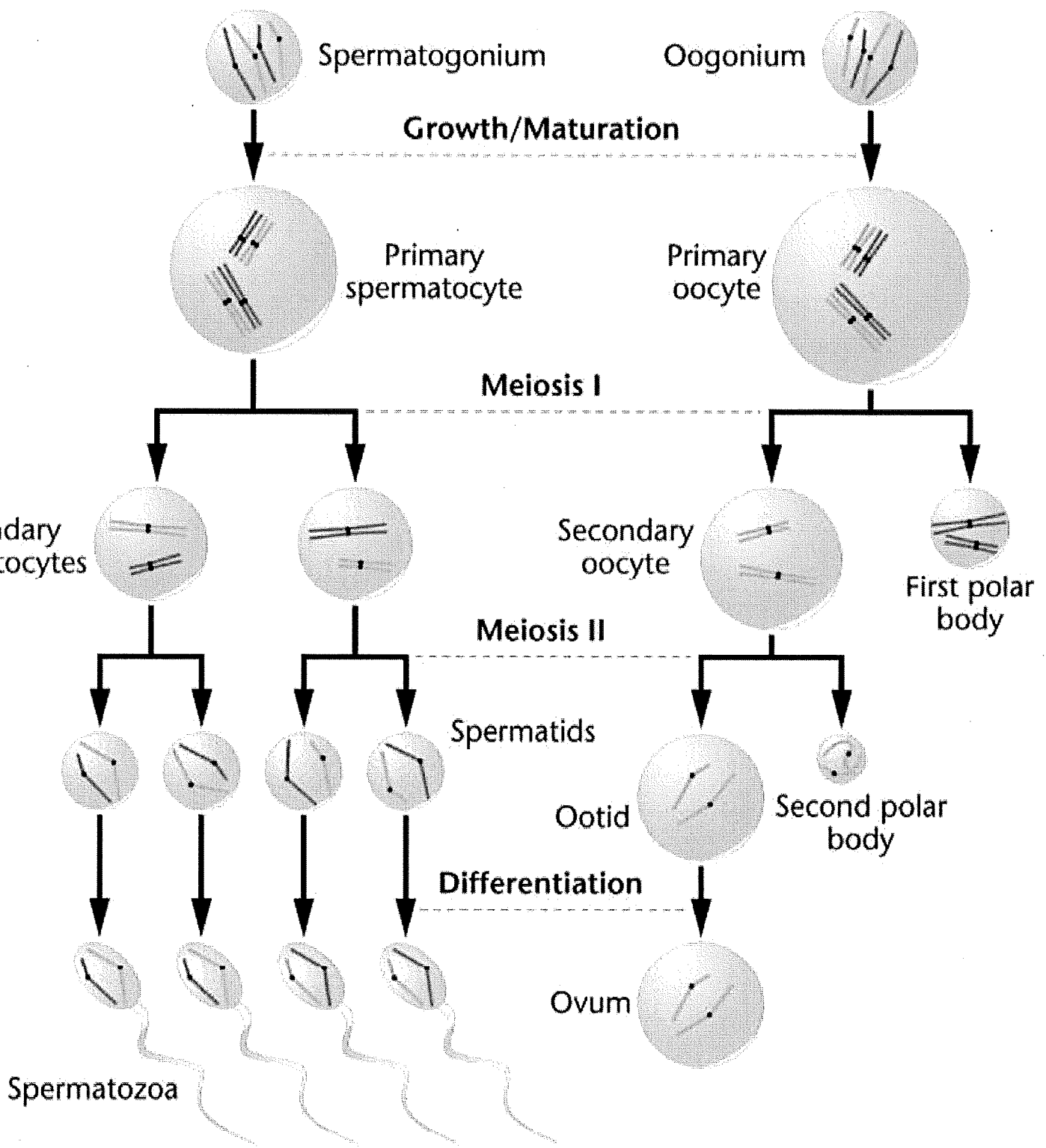
We are all born with DNA mutations and sometimes these can be beneficial or cause no problem.

When a gene variation is present in egg or sperm cells, it can be passed on to children (inherited).

In other cases, a new gene variation can arise in an egg or sperm cell. This is called a **de novo** change.

The person arising from that egg or sperm cell will be the first in the family to have the DNA change which may then be passed down to his or her children and future generations.

*Genes contain recipes for the body to make proteins - the Book of Life
is like a recipe book for our bodies*



2.2 Structure of DNA

Science understanding

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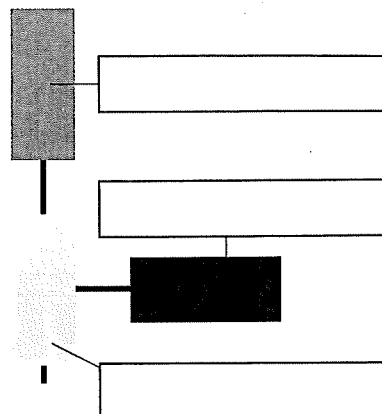
ADVANCED

DNA building blocks

The building blocks of a DNA molecule are called nucleotides. Each nucleotide has three parts:

- phosphate group
- sugar
- nitrogen-rich bases.

- 1 Use the three terms above to label the nucleotide diagram shown on the right.



DNA structure

The nitrogen-rich bases can be one of four types:

- adenine (A)
- thymine (T)
- cytosine (C)
- guanine (G)

In a double-stranded DNA molecule, A and T always form a pair and C and G always form a pair. They are called complementary pairs.

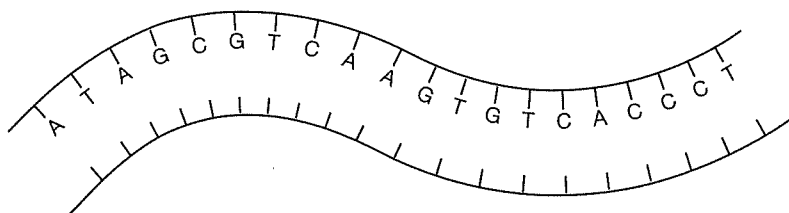
complementary (*adj*)
matching, one of a pair

DNA (*n*)
deoxyribonucleic acid—main part of chromosomes; transfers genetic characteristics

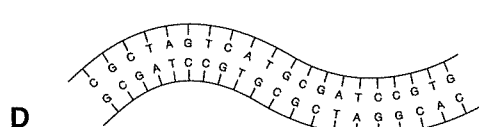
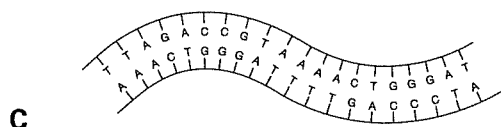
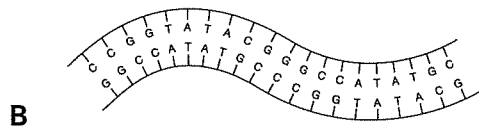
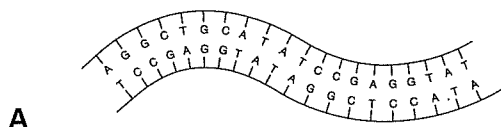
double-stranded (*adj*)
with two strands

strand (*n*) rope or thread

- 2 The sequence of bases below represents a section of a single strand of DNA. Write the base sequence in the complementary strand of DNA.



- 3 (a) Identify which of the following figures A, B, C, or D represents a possible correct base sequence in a molecule of DNA. _____



- (b) Justify your decision for each molecule, explaining why it is correct or incorrect.

- A** _____
- B** _____
- C** _____
- D** _____

2.3 Complementary base pairing

Science understanding

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The chemical structure of the nitrogen-rich bases means that they can only form chemical bonds with one of the other bases.

- Adenine (A) only pairs with thymine (T).
- Cytosine (C) only pairs with guanine (G).

One side of the DNA 'ladder' could be like Figure 2.3.1 with the sugar-phosphate backbone and the attached bases.

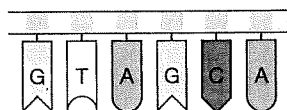


Figure 2.3.1 One side of the DNA strand

Using complementary base pairing, the other side of the molecule would look like Figure 2.3.2.

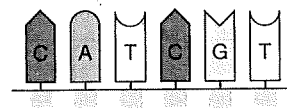


Figure 2.3.2 The other side of the DNA ladder

When the two sides are put together, the DNA molecule in Figure 2.3.3 would result.

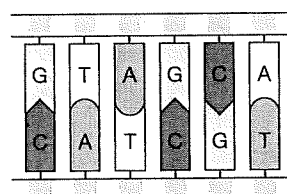
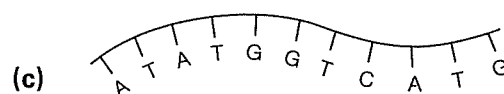
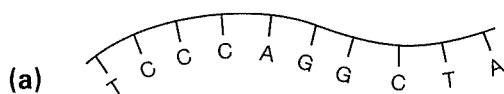


Figure 2.3.3 The resulting DNA molecule

- 1 Construct a diagram of the complementary DNA ladder for the following sequences.



2.4 Mitosis

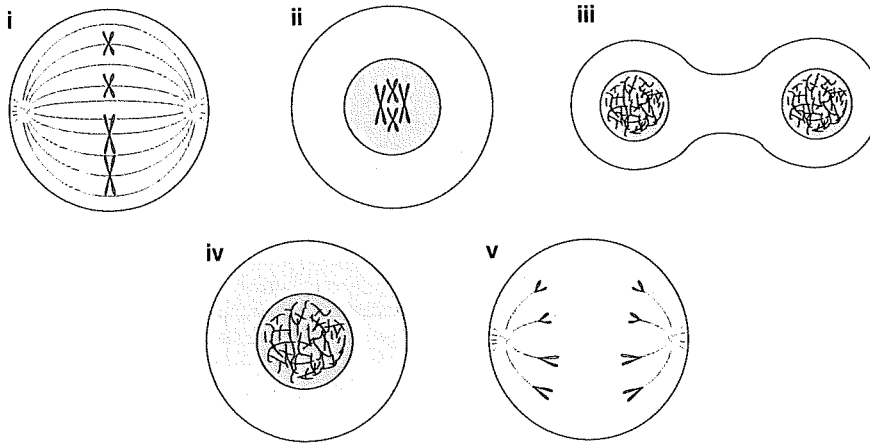
Science understanding

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The diagram represents five stages of mitosis. However, they are not in the correct order.



The following captions (A to E) are descriptions of the five stages of mitosis shown above. The descriptions are not in the correct order.

- A The chromatids separate. The spindle fibres contract, pulling the chromosomes to opposite poles of the cell.
- B Separate chromosomes become visible. Each chromosome comprises two chromatids.
- C In the period between cell divisions, the DNA replicates.
- D The nuclear membrane re-forms, enclosing the chromosomes into a new nucleus at each pole. Division of the nucleus is now complete. The cytoplasm then divides, resulting in two identical daughter cells.
- E The membrane surrounding the nucleus breaks down. The spindle appears, extending from the poles of the cell to each chromosome. The chromosomes line up across the equator of the cell.

comprise (v) made up of, to consist of
contract (v) to shorten
divide (v) to split in half
equator (n) middle, centre
identical (adj) exactly the same
replicate (v) to reproduce or copy itself

- 1 Use the table below to redraw each stage of mitosis in the correct order.
- 2 Identify the letter of the correct caption for each stage.

First stage	Second stage	Third stage	Fourth stage	Fifth stage
Caption:	Caption:	Caption:	Caption:	Caption:

2.5 Meiosis

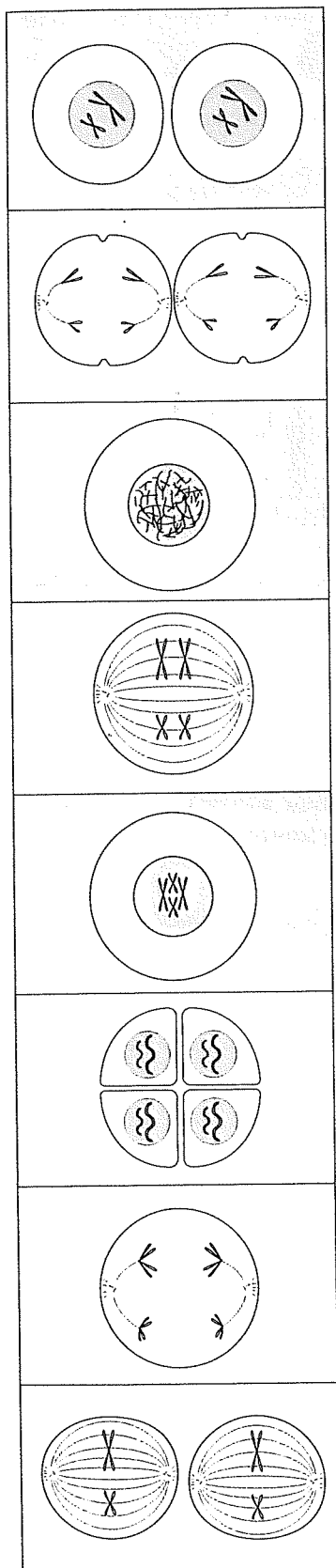
Science understanding

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- 1 Meiosis takes place in 8 different stages. Draw lines to match each diagram of meiosis on the right with its correct description on the left.



The nuclear membrane breaks down and pairs of replicated chromosomes line up on the equator of the cell with spindle fibres attached.

The nuclear membranes form and the cytoplasm divides to produce four new cells. Each cell now contains the haploid number of chromosomes. These cells are the gametes.

In the period between cell divisions, the DNA replicates.

The spindle fibres contract, pulling the chromatids apart towards the poles of the cells.

At the end of the first part of the division, nuclear membranes may re-form. The chromosomes at each pole are enclosed into new nuclei.

The spindle fibres contract, drawing one chromosome from each pair to opposite poles of the cell. At this stage, each chromosome is still two chromatids.

The DNA becomes visible as separate chromosomes, each of which comprises two chromatids.

After the first part of the division, a new spindle forms at right angles to the first. The new spindle fibres attach to the chromosomes and line them up on the equator of the cell.



Main Ideas

Copy and complete these statements to make a summary of this chapter. The missing words are on the right.

- _____ are found in the nuclei of cells. They carry _____ which determine what an organism looks like and how it functions.
- Body cells contain pairs of chromosomes and _____ contain only single chromosomes. So when fertilisation occurs, the single chromosomes in the sex cells form pairs in the cells of the new organism.
- Sex in humans is determined by sex chromosomes—females have a pair of _____ and males have an X and a _____.
- Organisms carry two genes for each characteristic. Some genes are _____ and mask the effects of the other (recessive) genes.
- The _____ and phenotypes of offspring can be predicted using _____.
- Chromosomes are made of _____. The sequence of the _____ on the DNA determines which types of proteins will be made by the cell.
- _____ alter the genetic material in cells, and occur spontaneously or from exposure to radiation or certain chemicals.
- Scientists throughout the world have collaborated to _____ all the genes in the human _____.

bases
chromosomes
DNA
dominant
genes
genome
genotypes
map
mutations
Punnett squares
sex cells
X chromosomes
Y chromosome

Test Yourself

- Chromosomes are found in—
 - sex cells only
 - all cells with a nucleus
 - fertilised eggs only
 - animal cells only
- A gene is—
 - a chromosome
 - a molecule of DNA
 - part of a chromosome that carries a single instruction
 - one base on a molecule of DNA
- A guinea pig has brown hair. Which one of the following statements is correct?
 - Both genes for hair colour came from the male parent.
 - Both genes came from the female parent.
 - One gene came from each parent.
 - Two genes came from each parent.
- Horses have a total of 64 chromosomes in each of their body cells. Male horses have an X and a Y chromosome in their cells.
 - How many pairs of chromosomes are found in horses?
 - How many chromosomes are found in the sperm of a horse?
 - Which chromosomes do sperm carry?

Review Questions

1. Complete the table to compare mitosis and meiosis in humans.

	MITOSIS	MEIOSIS
Function		
Location		
Chromosome number in parent cell		
Chromosome number in daughter cells		
Number of daughter cells		
Number of divisions		

2. (i) When meiosis occurs in the testes what proportion of the gametes formed contain an X chromosome?
- _____
- (ii) In the testes what proportion of gametes formed contain a Y chromosome?
- _____
- (iii) When meiosis occurs in the ovaries what proportion of the gametes contain an X chromosome?
- _____
- (iv) How does this affect the proportion of males and females in the population?
- _____

12.2 MEIOSIS



Terminology

These are some of the terms from this section which you should know. Write the meaning of each term in the space provided.

(i) cell division

(ii) chromatid

(iii) ~~chromatin~~

(iv) ~~cytokinesis~~

(v) daughter cell

(vi) homologous chromosomes

Review Questions

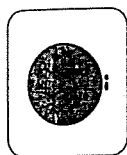
1. (i) Define haploid.

(ii) Which cells are haploid?

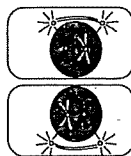
(iii) Define diploid.

(iv) Which cells are diploid?

2. (i) The following sequence shows a cell undergoing meiosis. This is not a human cell. In the spaces provided, describe what is occurring in and to the cell.



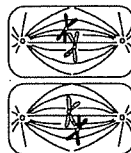
1. _____



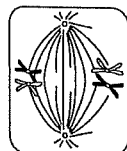
5. _____



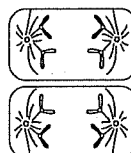
2. _____



6. _____



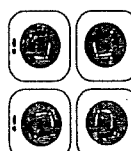
3. _____



7. _____



4. _____



8. _____

In the example above:

(ii) How many chromosomes did the parent cell contain? _____

(iii) How many chromosomes do each of the gametes contain? _____

(iv) What is the total number of chromosomes in the four daughter cells? _____

(v) How do you account for the apparent increase in the total number of chromosomes? _____

Name: _____

Date: _____

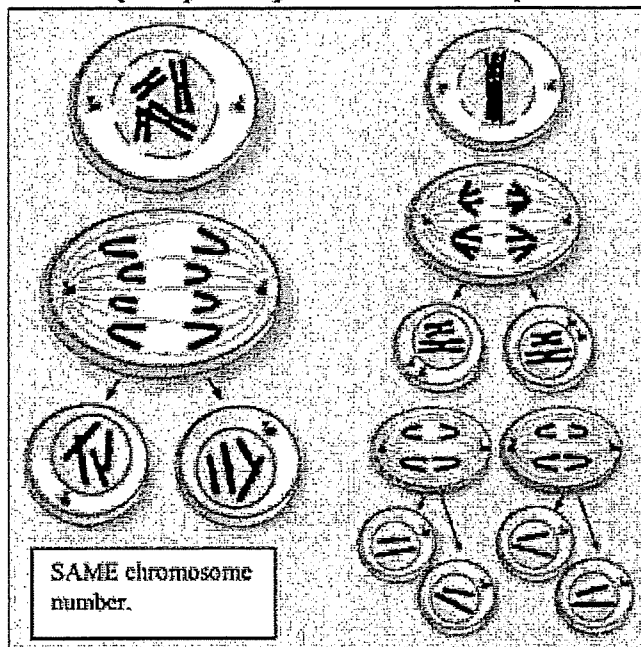
Period: _____

Comparing Mitosis & Meiosis Key

Directions: Use the image below to help you determine whether each statement describes mitosis, meiosis, or both. Place an "x" in the correct box.

Mitosis (body cells)

Meiosis (Sex Cells)

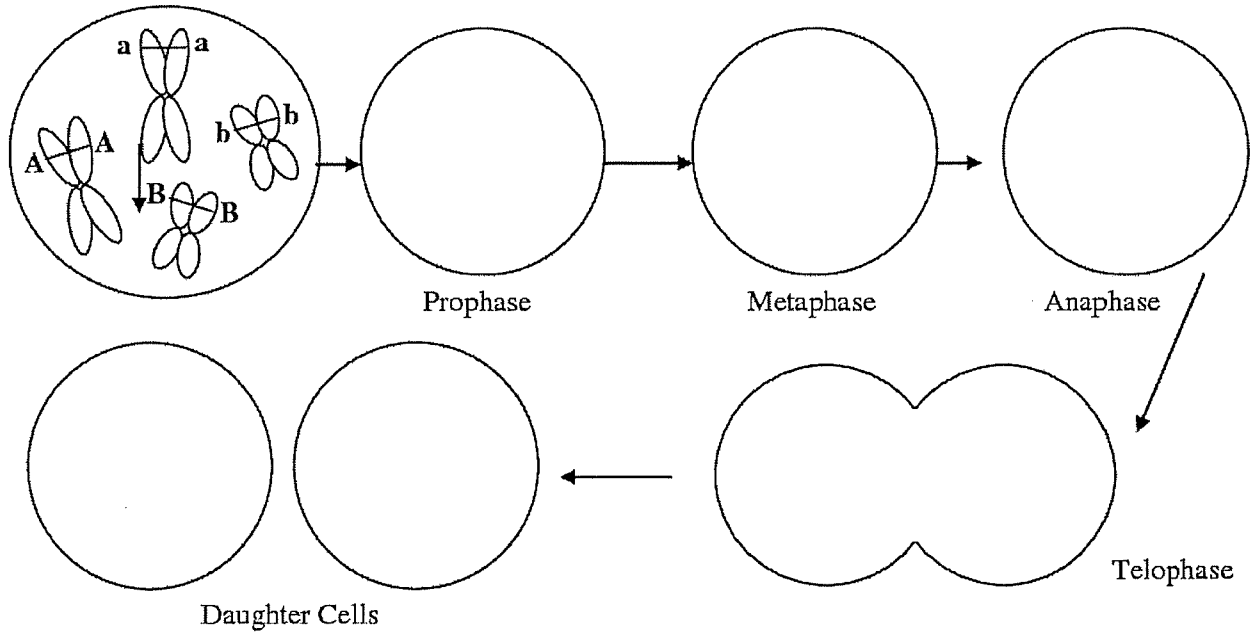


	MITOSIS	MEIOSIS	BOTH
1. There are two rounds of division in the process.			
2. Four total cells are produced.			
3. Duplicated chromosomes separate during the process.			
4. DNA is copied during interphase.			
5. At the end, the chromosome number is the SAME as the original cell.			
6. The process produces body cells.			
7. The process produces cells that are the SAME as the original cell.			
8. The process produces cells that are DIFFERENT from the original cell.			
9. The process produces sex cells (sperm & egg).			
10. The process produces cells with HALF the number of chromosomes.			
11. The process has only one round of division.			
12. Chromosome pairs separate during the process.			
13. Two total cells are produced.			

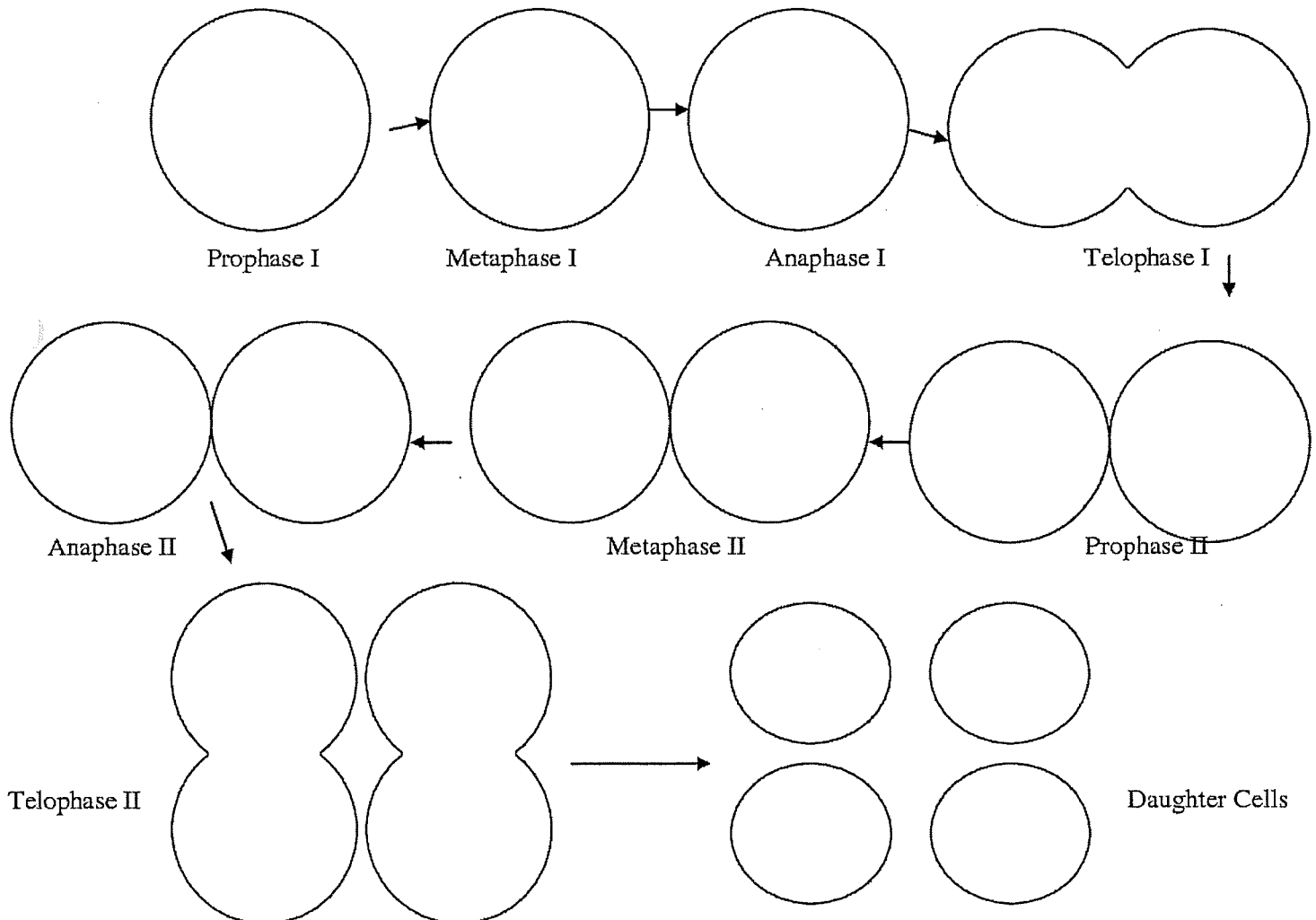
Assignment_Science7

Mitosis and Meiosis Worksheet
BioSci 110, Sections 20 and 28

I. Draw the chromosomes in the cell as it undergoes **Mitosis**:



II. Draw the chromosomes in the cell as it undergoes **Meiosis** (same parent cell):





Name: _____

Skill: knowledge

Mitosis

Cut out the descriptions and diagrams for the five stages of mitosis and arrange them in the correct order on page 2 of this worksheet.

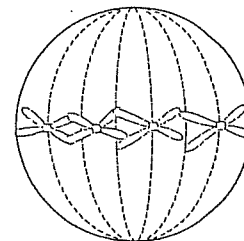
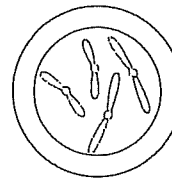
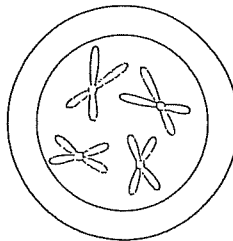
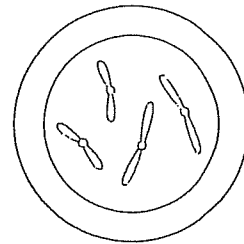
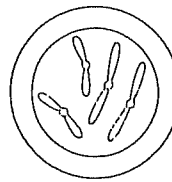
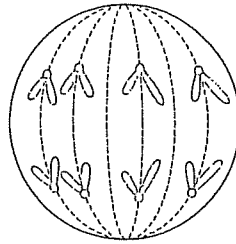
Chromosomes separate and move to the ends of the cell.

Two pairs of chromosomes are visible.

Chromosomes are doubled but attached at a point called the centromere.

Membranes form to produce two daughter cells.

Chromosomes line up along the 'equator' of the cell.



Meiosis

Cut out the descriptions and diagrams for the seven stages of meiosis and arrange them in the correct order on page 3 of this worksheet.

Chromosomes are doubled but attached at a point called the centromere.

Chromosomes separate and move to the ends of each cell.

Membranes form to produce two daughter cells.

Chromosomes line up along the 'equator' of the cell.

Homologous chromosomes line up along the 'equator' of the cell.

One of each pair of chromosomes moves to the ends of the cell.

Two pairs of chromosomes are visible.

